07 — Determinism

Deterministic vs Non-Deterministic Models

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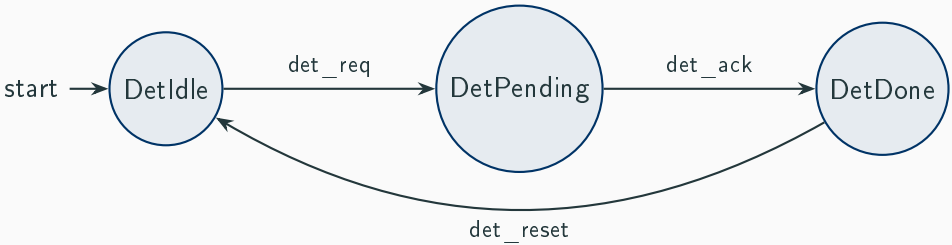
Deterministic Protocol

In a **deterministic** automaton, each (state, label) pair has exactly one target:

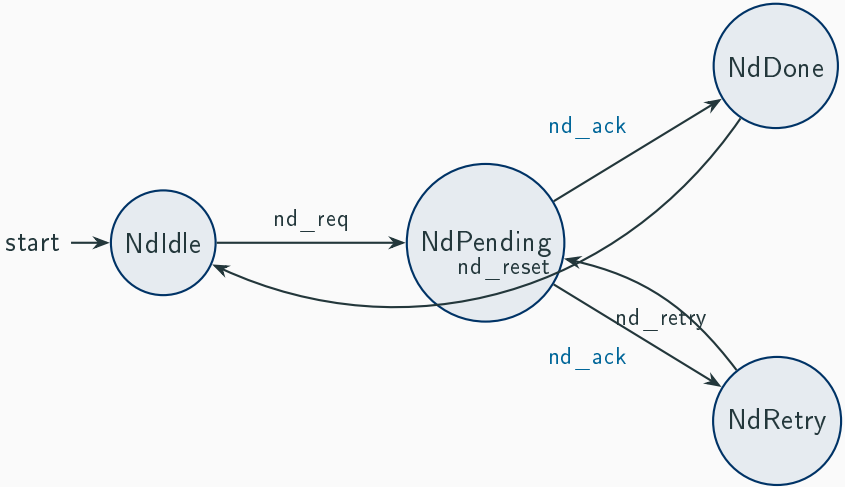


From **DetPending**, `det_ack` leads to *exactly one* state: **DetDone**.

No ambiguity — the next state is fully determined by the current state and the label.

Non-Deterministic Protocol

Now `nd_ack` from **NdPending** leads to two different states:



Impact on Verification: Box vs Diamond

Non-determinism changes the meaning of modalities:

□ — **Box (universal):** "*All* successors satisfy φ"

> $[\text{labels} = \{\texttt{nd\_ack}\}]$ *NdDone* is FALSE for NdPending
> because one `nd_ack`-successor is NdRetry, not NdDone.

◇ — **Diamond (existential):** "*Some* successor satisfies φ"

> $\langle \text{labels} = \{\texttt{nd\_ack}\} \rangle$ *NdDone* is **TRUE** for NdPending
> because one `nd_ack`-successor *is* NdDone.

In a **deterministic** model, box and diamond over a label agree (at most one successor).

Non-Deterministic Transitions in CTXDSL

```
1            }
2
3            transitions {
4                transition NdIdle -> NdPending on label nd_req;
5                // Non-deterministic choice: nd_ack leads to Done OR Retry
6                transition NdPending -> NdDone on label nd_ack;
7                transition NdPending -> NdRetry on label nd_ack;
8                // Retry loops back to try again
9                transition NdRetry -> NdPending on label nd_retry;
10               transition NdDone -> NdIdle on label nd_reset;
11           }
12       }
```

Lines 59–60: two transitions from **NdPending** on the *same* label `nd_ack`.

Formulas: Box vs Diamond

```
1        // Box modality: from NdPending, ALL nd_ack transitions lead to NdDone?
2        // This will be FALSE because nd_ack also leads to NdRetry
3        formula nd_ack_always_done {
4            over NondeterministicProtocol;
5            body = (! NdPending) || [ labels = { nd_ack } ] NdDone;
6        }
7
8        // Diamond modality: from NdPending, SOME nd_ack transition leads to
             NdDone?
9        // This will be TRUE (one branch does lead to NdDone)
10       formula nd_ack_some_done {
11           over NondeterministicProtocol;
12           body = (! NdPending) || < labels = { nd_ack } > NdDone;
13       }
```

- `nd_ack_always_done` uses [ ] (box) — requires *all* nd_ack-successors to be NdDone. False for NdPending.

Takeaway

Concept	Meaning
Deterministic	Each (state, label) has at most one target. Fully predictable.
Non-deterministic	Same (state, label) can have multiple targets. Models uncertainty.
Box $[\,]\,\varphi$	Universal — *all* successors satisfy φ.
Diamond $\langle\,\rangle\,\varphi$	Existential — *some* successor satisfies φ.

Key insight: Non-determinism is not a bug — it models environments whose behavior we cannot predict. Choose box or diamond depending on whether you need *guarantees for all* or *existence of some* outcome.